celebrated big-shot astronomer Sir Arthur Eddington,[fn13] whose work Chandrasekhar had refined, was not pleased about being outshone by this upstart, and made the young physicist's life miserable for years before eventually giving in to superior calculational excellence.)

COSMIC POPCORN

The idea that all white dwarf stars explode when they gather enough mass to exceed the Chandrasekhar Limit gives astronomers hope that we can use these stars as distance benchmarks, with some tweaks to account for slight differences in stellar circumstances.

Exactly how well we can do this is still a matter of incredibly intense debate in the astrophysics community. Which is understandable, as the stakes couldn't be higher. Type Ia supernovae are the gold standard[fn14] for distance measurements across vast expanses of the cosmos. They're what allowed astronomers in the late 1990s to detect the accelerated expansion of the universe, and they're what astronomers now use as their best handle on the nature of dark energy.

(It might sound odd to use massive stellar explosions as distance benchmarks, because, of course, we can't predict exactly when or where one will go off. But it turns out that the stellar explosion rate is high enough—a good rule of thumb is one supernova per galaxy per century—and there are so many galaxies, that if we just take pictures of lots of galaxies every night, we're likely pretty often to see a blip in one that wasn't there the night before, and then we can follow it up with more detailed observations.)

The precision with which we can now calibrate galaxy distances with supernovae is impressive, with accuracies pushing toward the 1 percent level. This makes it possible to measure the expansion rate of the universe, by determining how distant the galaxies are and how fast they're moving away. As discussed in Chapter 3, we talk about the expansion rate in terms of the Hubble Constant—the number that relates distance and recession speed. As of this writing, supernova measurements allow us to measure the Hubble Constant to an accuracy of 2.4 percent.

Which is weird, because the number we get totally disagrees with the value of the exact same number we derive from looking at the cosmic microwave background.